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SUPERSEDING
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17 April 1970

MILITARY SPECIFICATION

REEL, FUELING HOSE; AMPHIBIOUS, 6-INCH

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a diesel engine powered hose reel and cable reel assembly for installation on any suitable utility craft, such as warping tug.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-C-271 - Chains and Attachments, Welded, Weldless, and Roller Chain.
TT-E-490 - Enamel, Silicone Alkyd Copolymer, Semigloss (for Exterior and Interior Non Residential Use)
TT-P-645 - Primer, Paint, Zinc Chromate, Alkyd Type.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-T-75 - Tape, Preservation and Sealing.

MILITARY

MIL-P-116 - Preservation - Packaging, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2040

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MIL-M-7298 - Manuals, Technical: Commercial Equipment.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
DOD-P-15328 - Primer (Wash), Pretreatment (Formula No. 117 for Metals) (Metric).
MIL-L-17672 - Lubricating Oil, Hydraulic and Light Turbine, Noncorrosive.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

DRAWINGS

BUREAU OF YARDS AND DOCKS (NAVAL FACILITIES ENGINEERING COMMAND)

600 GPM Fuel System - Buoyant - Hose Reel:

993786 - Misc Details.
993788 - Assembly.
993789 - Reel Assembly.
993790 - Reel Details.
993791 - Cable Drum Assembly & Details.
993792 - Main Frame Assembly & Details.
993793 - Brake Supports & Shaft Clamps.
993794 - Misc Details.
993795 - Misc Details.
993796 - Hydraulic Control System.
1085655 - Protective Frame Assembly & Details.

(Copies of specifications, standards, and drawings required by supplies in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.
A176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip.

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A325 - Specification for High-Strength Bolts for Structural Steel Joints.
A563 - Carbon and Alloy Steel Nuts.
F436 - Hardened Steel Washers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19102.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS - D14.3 Specification for Welding Earth Moving and Construction Equipment.

(Application for copies should be addressed to the American Welding Society, 2501 NW 7th St., Miami, FL 33125.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 385 - Tank Vehicles For Flammable and Combustible Liquids.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J412 - Information Report for General Characteristics and Heat Treatments of Steel.
J514 - Standard for Hydraulic Tube Fittings.
J516 - Standard for Hydraulic Hose Fittings.
J517 - Standard for Hydraulic Hose.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Description. The reel assembly shall include a frame and base, a large reel having an 11-foot 8-inch diameter rim and a capacity for 5,000 feet of 6 inch collapsible hose, (and when specified (see 6.2.1) a small reel having a capacity for 5,000 feet of 1/2 inch diameter wire rope), a diesel engine and accessories, gear reducers, clutches, drive and control mechanisms, hydraulic pumps and motors, hydraulic fluid, braking systems, guards, spreader bars, lifting slings, protective frame, and other parts and components, all as specified herein, and on Drawings 993786, 993788 through 993796, and 1085655. The complete hose reel and component parts shall be herein after referred to as "unit". The unit shall operate in temperatures from +120° Fahrenheit (F) to -20°F.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish a complete unit for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extend possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3.1 Structural steel. Structural steel plates, bars, and shapes shall conform to ASTM Designation A36, except where otherwise shown on the applicable drawings.

3.3.2 Pipe. Unless otherwise specified (see 6.2.1), pipe shall be standard weight, black, electric resistance welded.

3.4 Construction. The entire unit shall have the dimensions, characteristics, and construction details specified herein and shown on Drawings 993786 and 993788 through 993796. The protective frame assembly shall be constructed as shown on Drawing 1085655. The contractor shall check all drawings and shall promptly notify the contracting officer of any discrepancies. The contractor shall compare all drawings and verify all figures before manufacture and shall be responsible for any errors which might have been avoided thereby. Dimensions, tolerances, and fits shall be uniform to provide interchangeability of like parts and components for all assemblies. All bearings shall be self-alining, precision, ball or roller type with self-alining grease seals. The engine base frame shall be of the slide rail type with cutouts for fork lifting. All drive mechanisms and brake systems, including clutches, belt and chain drive tension, and engine governor, shall be provided with adjusting devices, where necessary, to

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provide adjustment for proper operating characteristics, and to compensate for wear. The unit shall be furnished complete and assembled. When the mode of transportation prohibits complete assembly, the protective frame may be furnished knocked down and the hose drum may be removed.

3.4.1 Conflict between documents. Where conflict exists between this specification and referenced drawings, the drawings take precedence.

3.4.2 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Fasteners. Bolts, nuts, capscrews, setscrews, washers, and other fasteners shall be standard commercial quality of the characteristics, dimensions, finish, and quantities shown on the applicable drawings. Bolts, shall conform to ASTM A325, nuts shall conform to ASTM A563, washers shall conform to ASTM F436.

3.6 Diesel engine. The unit shall be powered by an aircooled, three cylinder, four cycle diesel engine, complete with all accessories and auxiliaries necessary for efficient and dependable operation. The power rating of the engine at a speed of 1,800 revolutions per minute (rpm) shall not be less than 33 continuous brake horsepower. The engine fuel filter, lubricating oil filter, air filter, and crankcase dipstick shall be easily accessible for maintenance. Unless otherwise specified, (see 6.2.1) the engine shall be provided with an "Ether Cold Start assistance" system for starting in cold climates. Unless otherwise specified (see 6.2.1), an electrical heating element in the crankcase oil pan for heating the oil shall be provided. The heating element shall heat the oil in the oil pan from -20°F to +20°F in a maximum time of 30 minutes. Power requirement shall be provided from the units batteries. The heating element shall be removable from outside the oil pan without removing the oil pan. The heating element shall not present a fire hazard in the oil pan while heating the oil.

3.6.1 Fuel system. The fuel system shall include a fuel filter, a pump, and a fuel tank of not less than 20-gallon capacity. The fuel tank shall be securely mounted to withstand vibration and shock loads encountered during operation and transportation. The fuel tank shall be located or protected to preclude fuel oil from contacting high temperature parts of the engine. A fuel gage shall be furnished on the tank.

3.6.2 Cranking system. An electric and hydraulic cranking system shall be furnished for cranking the engine. Both systems shall be capable of operating simultaneously and separately.

3.6.2.1 Electric cranking system. The engine shall have an electric cranking system based on a 24 volt circuit energized by wet storage battery(ies). Batteries shall be furnished dry and charged, without a filling of electrolyte. A keyless starter switch shall be mounted convenient to the operator. The system shall make a minimum of 3 starts of the engine at -20°F ambient. Switches and wiring shall meet requirements of NFPA 385 chapter 4.

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3.6.2.2 Hydraulic cranking system. The engine shall have a hydraulic system, consisting of a hydraulic cranking motor, two piston-type accumulators, an oil reservoir, an engine-driven pump, a hand-operated pump, a high pressure filter, and all connecting lines, isolating valves, and fittings to make a complete system and shall make at least one start with one charge at -20°F ambient.

3.6.3 Electric charging system. An electric charging system shall be provided for charging the 24 volt battery system.

3.6.3.1 Generator. The generator shall meet the requirements of NFPA 385 (see 6.4).

3.6.3.2 Wiring and main switches. The wiring and main switches shall meet the requirements of NFPA 385 (see 6.4).

3.6.4 Control system. The following controls shall be provided for the engine:

(a) Low lubricating oil pressure safety control of the shutoff type.
(b) A manually operated throttle control.
(c) A starter control lever mounted convenient to the operator.
(d) Main disconnect switch for the electrical starting system.
(e) Keyless starter switch.
(f) A quadrant type manually operated throttle, with detents for idle and for 1,800 rpm normal load speed, and with spring loaded detents for maximum recommended rpm and for stop positions, shall be mounted on the engine. A friction locking device shall be provided on the quadrant throttle. A remote throttle control, capable of being locked in any desired setting, and a flexible throttle cable suitable for operation in salt spray shall be provided.

3.7 Instrument panel. The operator's control position shall be provided with an instrument panel equipped with instruments in full view of, and for convenient operation by, the operator. All instruments shall be labeled to indicate their function. The instrument panel shall be equipped with, but not limited to, the following instruments:

(a) An engine tachometer.
(b) A cooling air outlet temperature gage and an ambient air temperature gage.
(c) An engine lubricating oil pressure gage and an oil temperature gage.
(d) An engine hour meter with totalizing mechanism of not less than 9,999 hours capacity. (This instrument may be mounted on the engine proper, in a visually accessible position.)
(e) A hydraulic fluid storage tank temperature gage.
(f) A hydraulic fluid pressure gage for the hose reel brake line.
(g) A hydraulic fluid pressure gage for the cable reel (when provided) brake line.
(h) A battery charging and voltage level meter.

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All instruments, as applicable, shall have color coded dials with amber to indicate below normal operating range, green for normal operating range, and red above normal or dangerous operating range. Instruments and gages exposed to the weather shall be weather proof type or have weather protective covers.

3.8 Hydraulic control system. The hydraulic control system shall consist of parts as shown on the drawings. The hydraulic control system shall meet all requirements specified herein using oil conforming to MIL-L-17672, symbol 2110 T-H. A suitable engine mounted double pump gear drive for mounting on the engine flywheel housing shall be used as shown on the applicable drawings. The double pump gear drive shall have proper gear ratios to satisfy the performance requirements for the unit including engine and pump speeds. The hydraulic control valves at the operator's position shall be variable flow type which will provide from 0.1 gallons per minute (gpm) to 10 gpm flow at 2,000 pounds per square inch (psi), and which will be suitable for the motors and load conditions.

3.8.1 Hydraulic pump. Two hydraulic pumps shall be furnished for activating the hydraulic control system. Each pump shall be of the inline piston type, capable of variable delivery, pressure compensated, 14 horsepower, with a capacity of 10 gpm at 2,000 psi. The pumps shall be of standard commercial quality. Each pump shall be provided with a 0 to 2,000 psi hydraulic pressure gage. The pressure compensators shall be set to a line pressure of 700 psi for the hose reel, and 1,200 psi for the cable reel.

3.8.2 Hydraulic motors. Hydraulic motors shall be of the piston type with a constant displacement, not less than 2.41 cubic inches per revolution, 760 inch-pound minimum torque, 2,000 minimum psi, and shall be of standard commercial quality.

3.8.3 Hydraulic tank. A hydraulic fluid storage tank shall be furnished as shown on the drawings, complete with fittings, 50 gpm oil strainer, oil level gage, oil filter, suction type, 10 micron (max) between reservoir and suction pump, and air breathing unit, and oil temperature gage sensor of standard commercial quality.

3.8.4 Tubing, valves, and fittings. All tubing, valves, and fittings necessary to complete the system, as shown on the drawings, shall be of standard commercial quality of the type best suited for the applicable pressures and service.

3.8.5 Pressure relief valve setting. The pressure relief valve shall be set at 2,100 psi.

3.9 Gear reducers. Gear reducers shall be foot mounted of the reduction ratio and type specified on the drawings. The gears shall be hobbed-type precision machine cut by controlled precision tooling, and shall be high strength, heat treated, ground, and polished steel. Shafting shall be high strength, ground, and polished steel. Antifriction bearings shall be high capacity ball or roller type. Housing shall be cast iron or welded steel of one or two sections, and shall be oiltight with double lip oil seals, filler and drain plugs, and a breather fitting. Lubrication system shall be the positive splash type supplying all movable parts with continuous flow of oil. The gear reducer shall be of standard commercial quality.

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3.10 Brakes. The brakes on the hose reel drive and cable reel (when provided) drive shall be constructed to conform to the drawings. Each brake unit shall be capable of developing a minimum torque of 42,500 inch-pounds at not less than 840 psi line pressure. Brakes for hose reel shall consist of two brake units acting on a 29 inch diameter disk plate. Brakes for cable reel shall consist of two brake units acting on a 19 inch diameter disk plate. Hydraulic lines for the brakes shall be routed in a manner which will give maximum protection against damage. Each brake unit, because of its peculiar orientation, shall be modified, as appropriate, to facilitate proper and complete bleeding of air from the brake system.

3.10.1 Brake plate. Brake plate(s) shall be fabricated from ASTM 176 405 stainless steel of the thickness shown on the drawings.

3.10.2 Brake supports and pillow blocks. Brake supports and pillow blocks shall be fabricated according to the drawings. Bolts indicated on the drawings shall be torqued as specified thereon. After assembly and careful alinement, the hose and cable reel pillow blocks shall be spot welded to the frame for rigidity, but permit subsequent realignment by removing spot welds.

3.11 Sprocket. The double row sprocket shall conform to dimensions shown on Drawing 993791, and shall be compatible with the roller chain.

3.12 Operator's platform. The operator's platform shall be constructed according to the drawings. A seat cushion and one seat belt of standard commercial quality shall be furnished.

3.13 Chain guards. Chain guards for the hose reel and cable reel (when provided) shall be constructed according to the drawings.

3.14 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Hydraulic lubrication fittings shall be located in accessible protected positions. Pressure lubrication fittings shall not be used where normal lubricating pressure may damage grease seals or other parts. The unit shall be lubricated prior to delivery. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.15 Chains and attachments. All chains and attachments shall be of the types indicated on the drawings and shall conform to RR-C-271, type III.

3.16 Bearings. Bearings for reels shall be alloy steel, straight bore, flanged cup, tapered roller bearings with self-alining, single collar, piloted flange mounting, and of the sizes shown on the drawings.

3.17 Synthetic rubber gaskets. Synthetic rubber gaskets shall be provided for leaktight joints, and shall be of the material, size, and configuration best suited for the intended use.

3.18 Flattened expanded metal. Flattened expanded metal shall be commercial grade, with openings and thickness as shown on the drawings.

3.19 Hydraulic hose. Hydraulic hose shall conform to SAE J517, for SAE 100R2 series hose.

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3.20 Hydraulic fittings. Hydraulic fittings shall be suitable for use with the hose described in 3.19. The fitting shall be of the type which do not require skiving of the hose to facilitate assembly. Female portions of the fittings shall be of the swivel nut type. The fitting shall be capable of holding beyond the burst pressure of the hose. All hydraulic fittings shall be of one manufacturer and shall conform to SAE J516, 37° flared, reusable type. Tubing fittings shall conform to SAE J514, 37° flared type.

3.21 Lifting and tiedown attachments. Lifting and tiedown attachments shall be provided as shown on the applicable drawings. The words TIEDOWN or LIFT shall be stenciled adjacent to each fitting, as appropriate. A diagram showing the lifting attachments and lifting sling and a silhouette of the equipment showing the center of gravity shall be provided. The lifting sling shall be as shown on the applicable drawings so as to provide a balanced level lift, except that the contractor shall determine the proper positioning of the intermediate pad eyes for lifting the hose reel drum and sprocket ring.

3.22 Treatment and painting.

3.22.1 Priming. Prior to assembly, surfaces of exposed ferrous metal parts (except engine and accessories, hydraulic pumps, motors, fittings, and brake discs and drums) shall be completely primed with zinc chromate primer conforming to TT-P-645 with a 2.0 mil minimum dry film thickness. Prior to priming, all rust, mill scale, corrosion, dirt, grease and oil shall be removed and surfaces shall be pretreated with a wash primer conforming to DOD-P-15328. Excepted items shall be treated and coated according to manufacturers standards.

3.22.2 Finish coating. After assembly, all welds shall be cleaned of slag, scale and spatter, all bare metal, bolts, screws, etc., shall be primed as above prior to application of the finish coat. All primed surfaces shall be furnished with two coats of enamel conforming to TT-E-490, color 24064 of Fed Std 595 to a 5 mils dry film, minimum. All paint shall be applied to dry surfaces in a dry atmosphere. If spray or roller painting is utilized, all welds and corners shall first be thoroughly brush coated. The finished paint application shall be smooth in appearance, continuous in color, free of dry overspray, pinholes, orange peel, sags, or other film defects.

3.22.3 Fungus resistance. Electrical components, wiring, and circuit elements, including terminal and circuit connections, shall be coated with varnish or other protective material as appropriate to prevent corrosion and other damage from moisture, other atmospheric elements, salt laden moisture, and mechanical damage according to good manufacturers standard practice, except that:

(a) Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
(b) Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.23 Safety. All rotating and reciprocating parts, or parts subject to high temperature, having inherent qualities or location that makes them hazardous to operating personnel shall be insulated, fully enclosed, or properly guarded. Guards shall be fabricated of solid or expanded sheet steel as shown on the drawings, and shall be firmly attached to the framework by means of bolts or screws.

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3.24 Identification.

3.24.1 Identification marking. Assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.24.2 Piece and matchmarking. Parts of the protective frame shall have identifying piece marks and matchmarks to simplify assembly.

3.25 Name and instruction plate.

3.25.1 Nameplate. A nameplate will be furnished by the contracting officer for each unit. The contractor shall stamp all necessary data in the blank spaces provided for that purpose and securely affix it to each unit in a conspicuous place with brass screws or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.25.2 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.25.3 Manuals. Unless otherwise specified (see 6.2.1), manufacturers manuals(s) shall be provided on the complete unit and component parts and assemblies. The manual(s) shall include: Instructions for assembly and disassembly from the shipping mode to the operation mode; operation, lubrication, adjustment procedures and techniques including types of fuel oils, lubricants, coolants required; winterization requirements for -20°F; cold start procedures and precautions. Parts and repair manual(s) shall be as specified (see 6.2.1).

3.26 Workmanship.

3.26.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.26.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice and all bolts, nuts, and screws shall be tight (see 3.5).

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3.26.3 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Welding shall be in accordance with AWS-D14.3.

3.26.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.26.5 Machine work. Quality of machine work shall meet or exceed the highest standard of the manufacturing industry. Where machined surfaces are required, adequate thicknesses of stock shall be used so that final thicknesses shall be as shown on the drawings, without holidays. Machining shall be performed to the sizes and dimensions shown on the drawings, making due allowances for heat treating, hardening, finish grinding, and plating or coating based on shop experience and judgement.

3.26.6 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for use.

3.26.7 Tolerances and shop practices. All exposed sheet metal edges shall be broken and returned. Unless otherwise shown on the drawings, fractional tolerances shall be 1/64 inch, angular tolerances shall be 1 minute, and three place decimals shall be 0.001 inch. Bearing mounting components shall be square and true within 90° $\pm$1 minute and flat within $\pm$0.002 inch.

3.27 Servicing and restoration. Each unit tested shall be serviced and restored to a service condition equal to the original condition of the unit, neglecting nominal wear incurred during the tests. The restoration shall include paint touchup or repainting, as required for delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification inspections. Inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. First article inspection shall be performed on one unit when a first article is required (see 3.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article shall be a sample unit produced prior to initiation of the production item. The test shall be performed in the maximum and minimum ambient temperatures specified in 3.1.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed as specified herein. This inspection shall include the examination 4.3 and the applicable tests of 4.4.

4.3 Examination. Each unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncomplance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. All tests shall be conducted with a 24,000 pound simulated hose load on the hose reel drum.

4.4.1 Lift tests. Lift tests shall apply to the completed first article, if furnished. Lift tests shall produce no breakage or permanent distortion of any member or component of the completely assembled unit.

4.4.1.1 Hose reel. The hose reel drum with attached sprocket ring shall be lifted with the sling to determine compliance with 3.21.

4.4.1.2 Assembled unit without frame. The complete unit, without the protective frame, shall be lifted by the sling to determine compliance with 3.21.

4.4.1.3 Assembled unit with frame. The complete unit with the protective frame attached, shall be lifted by the sling, using the lifting eyes of the protective frame, to determine compliance with 3.21.

4.4.2 Operational tests. Operational tests shall apply to the first article and when specified (see 6.2.1), to all subsequent production units offered by the contractor. The operational tests shall include but not be limited to those specified herein. The hydraulic fluid used for operational tests shall be steam turbine oil, conforming to MIL-L-17672, symbol 2110T-H. Failure of any component to meet requirements of section 3, as demonstrated by the appropriate tests, shall be cause for rejection of the entire unit. After repair, replacement, or redesign and replacement of the defective component, it shall be retested for its function in the unit. Operational tests shall be made in the maximum and minimum temperatures of 3.1 after the unit temperature has stabilized.

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4.4.2.1 Engine system, hydraulic start.

(a) Prior to engine starting, check adequacy of the following:

(1) Fuel level.
(2) Crankcase oil level.
(3) Air cleaner.
(4) Battery charge level.

(b) Prime fuel and oil systems in accordance with operating manual.
(c) Determine hydraulic system fluid level (engine starter) and system pressure and if system pressure is below 2,000 psi, pressurized with hand pump.
(d) Start engine in accordance with operating manual. Allow engine to warm up for a minimum of 5 minutes and check adequacy of the following:

(1) Engine oil temperature.
(2) Oil pressure.
(3) Generator output.

(e) Test oil heater system to determine compliance with 3.6.

4.4.2.2 Engine system, electric start.

(a) Perform prestart check specified in 4.4.2.1(a).
(b) Prime as specified in operators manual.
(c) Start engine with electric starter as specified in operators manual.
(d) Make after start check specified in 4.4.2.1(d).

4.4.2.3 Hydraulic system.

(a) Pressurize each brake system with the brake pedal to a minimum of 700 psi line pressure and check for leaks.
(b) With engine running and at operating temperature, perform the following check on the hydraulic system.

(1) Open hose reel control valve and apply hose reel brake.
(2) Set pressure compensator on hose reel to 700 psi system pressure.
(3) Open cable reel (when provided) control valve and apply cable brake.
(4) Set pressure compensator on cable reel pump to 1,200 psi system pressure.
(5) Check entire hydraulic system for leakage.

4.4.2.4 Reel mechanism and brake system.

(a) With engine running at operating temperature, perform the following on the cable reel (when provided) and hose reel mechanism.

(1) Open cable reel control valve wide open in the forward direction.

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(2) With cable reel control valve in the wide open position, open hose reel control valve wide open in the forward direction.
(3) Check hose reel and cable reel mechanism for any evidence of binding during operation.
(4) Repeat 1 through 3 in the reverse direction and check for any evidence of binding.

(b) With engine running and at operating temperature perform the following on the reel assemblies:

Cable reel (when provided).

(1) Open the cable reel control valve wide open in the forward direction.
(2) Apply the cable reel brakes.
(3) Check brake system pressure (700 psi).
(4) Check hydraulic system pressure (must be a minimum of 1,200 psi).
(5) Hold brakes for 2 minutes and check for slippage.
(6) Repeat 1 through 5 in reverse reel direction.

Hose reel.

(1) Open the hose reel control valve wide open in the forward direction.
(2) Apply the hose reel brakes.
(3) Check brake system pressure (700 psi).
(4) Check hydraulic system pressure (must be a minimum of 700 psi).
(5) Hold brakes for 2 minutes and check for slippage.
(6) Repeat 1 through 5 in reverse direction.

4.5 Packaging inspection. The preservation, packing, and marking of the unit shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. The preservation shall be level A or commercial, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish reduction in cube. Bolts, nuts, screws, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss.

5.1.1.2 Preservatives. Preservatives shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unpainted surfaces. Unpainted and uncoated exterior, ferrous metal surfaces, including threaded surfaces and surfaces exposed by disassembly, shall be coated with type P-1 preservative.

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5.1.1.4 Engines. Engines, component parts, and accessories shall be preserved in accordance with MIL-E-10062, type I, level A.

5.1.1.5 Disk-type clutches. Surfaces of clutch assemblies, such as clutch pressure plates, springs, exposed surfaces of disk hubs and spline shafts, linkage, pins, arms, sleeves, throwout collars, and all metal facings within the housings, shall be sprayed or fogged with a thin film of primer conforming to TT-P-664. Coating of composition facings shall be held to a minimum. Unpainted surfaces of control mechanisms not enclosed shall be coated with type P-1 preservative.

5.1.1.6 Enclosed gears. Gear housings shall be filled to the operating level with lubricant conforming to the applicable lubrication order. The gears shall be actuated to insure coating of all interior parts and surfaces.

5.1.1.7 Exposed drive chains. Drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains and sprockets shall be coated with type P-1 preservative.

5.1.1.8 Drive belts and pulleys. Belts shall either be released from tension or shall be completely removed. Unpainted faces of pulley grooves shall be coated with primer conforming to TT-P-664. Belts removed shall be preserved in accordance with MIL-P-116, method IC-3.

5.1.1.9 Hydraulic systems. The hydraulic fluid supply tank shall be filled to the operating level with approved hydraulic fluid. Exposed surfaces of the hydraulic pump shaft shall be coated with type P-11, preservative. The preserved surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type II, grade A, class 2, secured in place with tape conforming to PPP-T-75, type II.

5.1.1.10 Openings. Vents into fuel and hydraulic tanks, openings into accumulators and hydraulic pump and other opening which will premit entry of moisture or foreign matter shall be sealed with tape conforming to PPP-T-75, type II. Large openings shall be covered with barrier material conforming to MIL-B-121, type II, grade A, class 2, secured in place with tape specified herein.

5.1.1.11 Instruments. The dial glass of instruments shall be covered individually or in groups with fitted piece of plywood secured with tape conforming to PPP-T-75, type II.

5.1.1.12 Repair parts. Repair parts shall be preserved and packaged in accordance with level A of MIL-R-196. For parts not specifically covered in MIL-R-196, the applicable methods of preservation in MIL-P-116 shall be used.

5.1.1.13 Maintenance tools and technical publications. Unprotected metal surfaces of maintenance tools shall be coated with type P-1 preservative. Technical publications for each reel assembly shall be preserved in accordance with MIL-P-116, or IC-3.

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5.1.2 Commercial. The complete reel assembly shall be preserved in accordance with the contractor's standard practice.

5.2 Packing. The packing shall be level A, or commercial, as specified (see 6.2.1).

5.2.1 Level A. The reel assembly shall be shipped uncrated. The reel shall be secured to prevent movement during shipment. Lifting cables shall be secured to the unit in a protected location. Disassembled components, requiring additional protection from mechanical damage, and the repair parts, maintenance parts, maintenance tools, and technical publications for each reel assembly shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type. The contents shall be cushioned, blocked, and braced to prevent movement inside the container or damaging of the contents. Where practicable, the disassembled components, repair parts, tools, and technical publications shall be positioned and secured to the reel assembly by bolting or with flat steel strapping.

5.2.2 Commercial. The complete reel assembly shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing and containers shall comply with Uniform Freight Classification or National Motor Freight Classification.

5.3 Marking. Interior packages, shipping containers, and the reel assembly shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The unit is part of a 6-inch diameter buoyant hose, ship-to-shore fuel delivery system used in conjunction with amphibious assault bulk fuel handling systems.

6.2 Ordering data.

6.2.1 Acquisition requirements.

(a) Title, number, and date of this specification.
(b) When cable reel is required (see 3.1).
(c) When first article is required (see 3.2).
(d) When pipe is other than specified (see 3.3.2).
(e) When Ether Cold Start assistance or heating element not required (see 3.6).
(f) Manuals required if different (see 3.25.3).
(g) When operational test required on subsequent production unit (see 4.4.2).
(h) Level of preservation and level of packing required (see 5.1 and 5.2).

6.2.2 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data

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Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete unit or it may be standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examination, tests, and approval of the first article.

6.4 Generator, wiring and main switch(s). NFPA 385 chapter 4 section 41 "Auxiliary Electric Generators and Motors" provides regulatory requirements as indicated in 3.6.3.1 and 3.6.3.2.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-28572A(YD) REEL, FUELING HOSE; AMPHIBIOUS, 6-INCH

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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